

Sriparvathi Shaji Bhattathiri

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PROFILE

Ph.D. candidate in Engineering at Rochester Institute of Technology working at the intersection of artificial intelligence, robotics, and simulation for intelligent automation. My research develops reinforcement-learning-based dispatching for fleets of autonomous mobile robots, digital twin and stochastic simulation frameworks for warehouse systems, and human-robot interaction systems (gesture recognition, intent communication) that improve efficiency, safety, and trust in shared workspaces. My work bridges industrial engineering, human factors, and autonomous systems to address real-world problems in logistics, manufacturing, and production. Expected graduation: December 2026.

EDUCATION

- **Rochester Institute of Technology** *Expected: Dec. 2026*
Rochester, New York, USA
Ph.D. in Engineering
 - GPA: 3.63/4.00
 - Dissertation: Smart Autonomous Mobile Robot (AMR) Dispatching and Human-Robot Interaction
 - Advisor: Dr. Michael E. Kuhl
- **Rochester Institute of Technology** *Aug. 2020*
Rochester, New York, USA
M.S. in Electrical Engineering
 - GPA: 3.58/4.00
 - Project: Detection of Mental State Using EEG Signals (Advisor: Dr. Ferat Sahin)
- **Birla Institute of Technology and Science, Pilani – Dubai Campus** *Aug. 2017*
Dubai, UAE
B.E. (Hons.) in Electronics and Instrumentation Engineering
 - GPA: 8.5/10.0
 - Projects: Gas Safety Device, Level Process Control (Advisors: Dr. Swarnalatha R, Dr. Kalaichelvi Venkatesan)

RESEARCH EXPERIENCE

Graduate Research Assistant, Toyota Production System Lab (Intelligent Material Handling Systems), RIT

- **Reinforcement Learning-Based Robot Dispatching for Intralogistics** *Jan. 2023 – Present*
Rochester, New York, USA
Doctoral research
 - Built a simulated warehouse environment with realistic material handling tasks
 - Developed a single-agent RL dispatching algorithm that delivers more tasks on time than rule-based dispatching across different due-date tightness levels
 - Developed a multi-agent (2–10 AMR) RL dispatching algorithm that delivers more tasks on time than rule-based dispatching across different due-date tightness levels
- **Framework for Human-Autonomous Mobile Robot Communication** *Feb. 2024 – Mar. 2025*
Rochester, New York, USA
Funded by Toyota
 - Created a gesture language for human-robot interaction (HRI) in warehouses
 - Developed a CNN-based gesture recognition algorithm with 97.5% test accuracy
 - Implemented a real-time system that triggers robot actions from detected gestures
- **Effective and Efficient Robot Interaction for Material Handling** *Apr. 2021 – May 2022*
Rochester, New York, USA
Funded by Toyota
 - Designed visual robot communication systems for warehouse HRI with three interfaces: text-based, symbol-based, and digital-avatar-based
 - Implemented human pose estimation using MediaPipe
 - Digital avatar ranked highest among the interfaces; achieved a 20% reduction in task time and a 25% improvement in worker trust and safety perception
- **Intelligent Material Handling (Robot Perception)** *Feb. 2020 – May 2020*
Rochester, New York, USA
Funded by Toyota
 - Integrated LiDAR and 3D depth cameras with ROS for autonomous driving
 - Implemented RGB image-based person and forklift detection
- **Volume Estimation of Food from 2D Images** *Jan. 2020 – May 2020*
Rochester, New York, USA
Collaboration with the University of Rochester
 - Developed computer vision techniques for food volume estimation using frame transformation and Structure from Motion (SfM)

- Used MeshLab to convert 3D point clouds to convex hulls for volume computation

TECHNICAL SKILLS

Languages: Python, C, C++, Java, MATLAB
ML & Robotics: PyTorch, TensorFlow, ROS, Gazebo, OpenAI Gym, MediaPipe
Methods: Deep and multi-agent reinforcement learning, computer vision, discrete-event simulation, digital twins, optimization

TEACHING EXPERIENCE

- **Primary Instructor** Aug. 2025 – May 2026
Engineering Economics, Kate Gleason College of Engineering Rochester, New York, USA
 - Developed course materials, schedule, syllabus, assignments, and assessments
 - Taught undergraduate engineering students for two semesters (approx. 25 students per class)
- **Graduate Teaching Assistant** Aug. 2020 – Present
Multiple engineering courses Rochester, New York, USA
 - Courses: Data Analytics and Predictive Modeling (Fall 2026); Production Planning (Fall 2026); Simulation Analysis (Spring 2024); System Simulation (Fall 2023, Fall 2022); Facilities Planning (Spring 2024, Spring 2023); Engineering Value Creation (Fall 2023); Special Topics: Product Line Management (Fall 2020)
 - Evaluated student work and gave feedback; developed lesson plans and led laboratory sessions; held office hours and proctored exams

INDUSTRY EXPERIENCE

- **Estimation Engineering Intern** Jan. 2018 – Jun. 2018
Aspire Controls and Solutions Dubai, UAE
 - Designed building management systems (BMS) and lighting control systems using Visio and AutoCAD; prepared BOMs and cost estimates
- **Engineering Intern** Jan. 2017 – Jul. 2017
Emerson Rosemount Dubai, UAE
 - Designed a hazardous-chemicals tracking system and trained instrument builders to use it
 - Developed status reports for invoice tracking and project progress; managed budgeting and documentation for spare parts
- **Pre-sales Engineering Intern** Jun. 2015 – Aug. 2015
Telematics Networking & Communications Dubai, UAE
 - Designed LAN cabling layouts; prepared BOMs and pre-sales cost estimates

PUBLICATIONS

J = Journal, C = Conference

- [J1] **S. S. Bhattathiri**, M. E. Kuhl, and E. Rashedi, "Robot Perception Framework for Gesture Recognition," 2026 (in preparation).
- [J2] **S. S. Bhattathiri**, B. Kazempour, L. Vredenburg, M. E. Kuhl, and E. Rashedi, "RoboGestDB: A Gesture Dataset for Human–Robot Interaction," 2026 (in preparation).
- [J3] B. Kazempour, **S. S. Bhattathiri**, M. E. Kuhl, and E. Rashedi, "Framework for Human–Robot Communication Gesture Design: A Warehouse Case Study," 2026 (under review, *International Journal of Industrial Ergonomics*).
- [J4] **S. S. Bhattathiri**, A. Bogovik, M. Abdollahi, C. Hochgraf, M. E. Kuhl, A. Ganguly, A. Kwasinski, and E. Rashedi, "Unlocking human–robot synergy: The power of intent communication in warehouse robotics," *Applied Ergonomics*, vol. 117, Art. 104248, 2024.
- [J5] K. Patel, V. Kalaichelvi, R. Karthikeyan, and **S. Bhattathiri**, "Modelling, simulation and control of incremental sheet metal forming process using CNC machine tool," *Procedia Manufacturing*, vol. 26, pp. 95–106, 2018.
- [C1] **S. S. Bhattathiri**, M. E. Kuhl, A. Tondwalkar, and A. Kwasinski, "Simulation Analysis of a Reinforcement-Learning-Based Warehouse Dispatching Method Considering Due Date and Travel Distance," in *Proc. 2023 Winter Simulation Conference (WSC)*, pp. 1830–1841, 2023.
- [C2] F. Hyder, **S. S. Bhattathiri**, B. Yan, and M. E. Kuhl, "Deep Learning-Based Transmission Line Screening for Unit Commitment," in *Proc. 2023 IEEE Power & Energy Society General Meeting (PESGM)*, pp. 1–5, 2023.
- [C3] **S. S. Bhattathiri**, A. Bogovik, M. Abdollahi, C. Hochgraf, M. E. Kuhl, A. Ganguly, et al., "Towards Improving the Human–Robot Communication in Intelligent Materials Handling Environments," in *Proc. IISE Annual Conference and Expo*, 2023.
- [C4] M. E. Kuhl, R. Bhisti, **S. S. Bhattathiri**, and M. P. Li, "Warehouse Digital Twin: Simulation Modeling and Analysis Techniques," in *Proc. 2022 Winter Simulation Conference (WSC)*, Singapore, pp. 2947–2956, 2022.
- [C5] **S. S. Bhattathiri**, M. P. Li, and M. E. Kuhl, "A Traffic Avoidance Path Planning Method for Autonomous

Vehicles in a Warehouse Environment,” in *Proc. 2021 Winter Simulation Conference (WSC)*, pp. 1–10, 2021.

PRESENTATIONS AND OUTREACH

- **Winter Simulation Conference 2023** (San Antonio, Texas, Dec. 2023)
 - Paper: “Simulation Analysis of Reinforcement Learning-based Warehouse Dispatching”; Poster: “Integrating AI and Simulation for Material Handling”
- **Winter Simulation Conference 2021** (Phoenix, Arizona, Dec. 2021)
 - Paper: “Traffic Avoidance Path Planning for Autonomous Vehicles”
- **Imagine RIT Festival** (annual innovation festival, 30,000+ attendees)
 - Demonstrated HRI research: Warehouse Gesture Language, Recognition and Robot Action (2025); Pose-based HRI (2024); Symbol-based Robot Communication System (2023); Robot Perception Systems and Pose Detection (2022); SLAM and Sensor Integration for Robot Perception (2021)

HONORS AND AWARDS

- **Special Mention, Simio Student Simulation Competition** *Dec. 2020*
Simulated the Simio BBQ Smoke Pit to maximize KPIs and reduce customer waiting time
- **Best Hardware Integration, WiCHacks Hackathon** *Mar. 2019*
Designed a query-based chatbot to help users find dishes and recipes by ingredient

PROFESSIONAL MEMBERSHIPS

INFORMS, IISE, IEEE, Sigma Xi

GRADUATE COURSEWORK

Deep Learning for Computer Vision, Deep Learning, Machine Learning, Artificial Intelligence, Advanced Robotics, Principles of Robotics, CNC and Robots, Bio-robotics and Cybernetics, System Modeling and Optimization, System Simulation, Random Signals and Noise, Advanced Engineering Mathematics, Engineering Analysis, Interdisciplinary Research Methods, Translating Discovery into Practice

REFERENCES

Available upon request.